(a) Offline training

MICROGRID PLANT (decision step 20 ms, physics 2 ms)

Measured weather (recorded levels)
BSRN Cabauw 1-Hz GHI · FINO1 80 m sonic wind (1 s)
chronological train / validation / test split

Solar PV 500 kW
$P_{pv} = f(G, T_{cell})$

Type-4 wind turbine 500 kW
rotor dynamics, MPPT, pitch

Load 1800 kW
+ two load contingencies
(randomised in training)

Grid-forming BESS 2500 kVA
1250 kWh; schedule (τ_{sch} = 1 s)
+ 10 % droop + VSG:
$p_{vsg} = -2J_{eff}\, d\Delta\omega/dt - D_{eff}\, \Delta\omega_m$

J_{eff}, D_{eff}

Converter headroom limiter (Eqs. 20–25)
$h_{req} = 0.08\ J + 0.01\ D$ vs $h_{avail} = (S - |P_{sched}|) / S$
if $h_{req} > h_{avail}$: $(J_{eff}, D_{eff}) = (J, D) \cdot h_{avail} / h_{req}$

Swing equation
H_{sys} = 0.5 s, f_o = 50 Hz
$\Delta P = P_{pv} + P_w + P_{bess} - P_{load}$

Voltage channel
Q–V droop K_q, X_{th} = 0.25 pu
PCC voltage V

PLL / measurement
τ = 20 ms, σ = 8 mHz → Δf_m, $RoCoF_m$, V, P, SOC, h_{avail}

Δf_m, $RoCoF_m$, h_{req}, h_{avail}, trip

action $a_t = [J, D, K_q]$
slew limit (full range 0.3 s)

state s_t

DDPG / TD3 AGENT

DDPG
one critic $Q_\phi(s, a)$
17→256→256→1
+ target critic

TD3
twin critics $Q_{\phi 1}$, $Q_{\phi 2}$
$y = r + \gamma \min(Q'_1, Q'_2)$
smoothing σ 0.1, clip 0.3
actor every 2 updates

minibatch

Prioritised replay buffer
SumTree, 2×10^5 (s, a, r, s'), α 0.6, β 0.4→1.0

Multi-objective reward r_t
frequency, RoCoF (500 ms) + 2 Hz/s hinge,
voltage, effort, SOC, BESS power,
reserve $0.5 \cdot h_{req}$, infeasibility, in-band, trip
learner reward $10^{-3} \cdot (r - 1)$

$\nabla_a Q$

Actor $\mu_\theta(s)$
14→256→256→3, LayerNorm + ReLU, tanh
Polyak targets τ = 0.005
+ exploration noise 0.30 → 0.03

State s_t (14 inputs)
Δf_m, $RoCoF_m$, t/T,
P_{pv}, P_w, v_w, G, SOC,
ΔV, P_{bess},
J, D, K_q (values in force),
h_{avail}

(b) Online deployment

INVERTER CONTROLLER (every 20 ms)

Measurements
PLL frequency and RoCoF, V, P, SOC, h_{avail}
→ state s_t (14)

Trained actor μ_θ (frozen)
best validated checkpoint
0.083 ms per 20 ms decision (one CPU core)

Slew-rate limiter
J, D, K_q: full range in 0.3 s

Headroom limiter (Eq. 24)
derate (J, D) to the available headroom

BESS inverter control
VSG (J_{eff}, D_{eff}) + droop + Q–V droop K_q
→ microgrid

plant response

Critics, replay buffer and reward are used only during offline training.